

Module 5: Multivariable Calculus

Instructional Hours: 4

1 Introduction to multivariate functions

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/TrcCbdWwCBc>

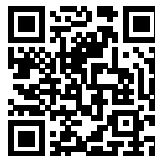


1.1 Definitions and Concepts

1.1.1 Finding the domain of multivariable functions

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/CxtgMNAQr8g>



Multivariable Calculus extends Single-Variable Calculus to functions of multiple variables. A multivariable function $f(x, y)$ depends on two or more variables. Its domain is the set of all valid (x, y) pairs, and its range is the set of all possible output values. The general form is:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

where n is the number of variables.

Domain: The set of all possible input values $(x_1, x_2, \dots, x_n)$ for which the function is defined. Range: The set of all possible output values.

Example 1. • *Function:* $f(x, y) = x^2 + y^2$

- *Domain:* All real numbers $\mathbb{R}^2$
- *Range:* $[0, \infty)$, since $x^2 \geq 0$ and $y^2 \geq 0$

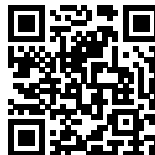
Example 2. • *Function:* $f(x, y) = e^x \sin(y)$

- *Domain:* All real numbers $\mathbb{R}^2$
- *Range:* $(-\infty, \infty)$ since e^x is positive and $\sin(y)$ ranges between -1 and 1 .

2 Partial Derivatives

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/CxtgMNAQr8g>



Partial derivatives measure how a function changes as one variable changes while keeping others constant. The partial derivative of a function $f(x, y)$ with respect to x measures how f changes as x changes, while y is held constant.

For a function $f(x, y)$:
 $\frac{\partial f}{\partial x}$ measures the rate of change of f with respect to x

$\frac{\partial f}{\partial y}$ measures the rate of change of f with respect to y

Example 3. • *Function:* $f(x, y) = x^2y + 3xy^2$

- *Partial Derivative with Respect to x :*

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^2y + 3xy^2) = 2xy + 3y^2$$

- *Partial Derivative with Respect to y :*

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^2y + 3xy^2) = x^2 + 6xy$$

Example 4. • *Function:* $f(x, y) = e^{xy}$

- *Partial Derivative with Respect to x :*

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} e^{xy} = ye^{xy}$$

- *Partial Derivative with Respect to y :*

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} e^{xy} = xe^{xy}$$

3 Gradients, Divergence, and Curl

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/ErZGbQeW1AQ>



Finding the gradient vector and the tangent plane of a multivariable function

3.1 Gradient

Definition: The gradient of a function $f(x, y)$ is a vector that points in the direction of the greatest rate of increase of f and has a magnitude equal to the rate of increase. It is a vector of its partial derivatives:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

3.1.1 Gradient Calculation

Example 5. • *Function:* $f(x, y) = x^2 + y^2$

• *Gradient:*

$$\nabla f = \left(\frac{\partial}{\partial x} (x^2 + y^2), \frac{\partial}{\partial y} (x^2 + y^2) \right) = (2x, 2y)$$

3.2 Vector Fields

Definition: A vector field assigns a vector to each point in a region of space.

- Notation: $F(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z))$
- Example: $F(x, y, z) = (x, y, z)$ represents a vector field where each vector points in the direction of the position vector.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/TCZ1GMoaUJw>



Finding the divergence and the curl of a vector field

3.3 Divergence

Divergence measures the rate at which a vector field spreads out from a point. For a vector field $\mathbf{F} = (F_1, F_2)$:

$$\operatorname{div} \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}$$

3.3.1 Divergence Calculation

Example 6. • *Vector Field:* $\mathbf{F} = (x^2y, 3xy)$

- *Divergence:*

$$\operatorname{div} \mathbf{F} = \frac{\partial (x^2y)}{\partial x} + \frac{\partial (3xy)}{\partial y} = 2xy + 3x$$

3.4 Curl

Curl measures the rotation of a vector field. For a vector field $\mathbf{F} = (F_1, F_2, F_3)$:

$$\operatorname{curl} \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

3.4.1 Curl Calculation

Example 7. *Problem:* Calculate the curl of the vector field $\mathbf{F} = (2xy, x^2 - y^2, z^2)$.

Solution: The curl of a vector field $\mathbf{F} = (F_1, F_2, F_3)$ is given by:

$$\nabla \times \mathbf{F}$$

where:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

- Write down the components of F :

$$\mathbf{F} = (2xy, x^2 - y^2, z^2)$$

where $F_1 = 2xy$, $F_2 = x^2 - y^2$, and $F_3 = z^2$.

- Set up the determinant for the curl calculation:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 - y^2 & z^2 \end{vmatrix}$$

- Compute the determinant: Expanding the determinant along the first row:

$$\nabla \times \mathbf{F} = \mathbf{i} \left(\frac{\partial}{\partial y} z^2 - \frac{\partial}{\partial z} (x^2 - y^2) \right) - \mathbf{j} \left(\frac{\partial}{\partial x} z^2 - \frac{\partial}{\partial z} (2xy) \right) + \mathbf{k} \left(\frac{\partial}{\partial x} (x^2 - y^2) - \frac{\partial}{\partial y} (2xy) \right)$$
- Calculate each component:

– For i :

$$\begin{aligned} \frac{\partial}{\partial y} z^2 &= 0 \quad (\text{since } z^2 \text{ does not depend on } y) \\ \frac{\partial}{\partial z} (x^2 - y^2) &= 0 \quad (\text{since } x^2 - y^2 \text{ does not depend on } z) \\ i(0 - 0) &= 0 \end{aligned}$$

– For j :

$$\begin{aligned} \frac{\partial}{\partial x} z^2 &= 0 \quad (\text{since } z^2 \text{ does not depend on } x) \\ \frac{\partial}{\partial z} (2xy) &= 0 \quad (\text{since } 2xy \text{ does not depend on } z) \\ -j(0 - 0) &= 0 \end{aligned}$$

– For k :

$$\begin{aligned} \frac{\partial}{\partial x} (x^2 - y^2) &= 2x \\ \frac{\partial}{\partial y} (2xy) &= 2x \\ k(2x - 2x) &= 0 \end{aligned}$$

- Combine the components:

$$\nabla \times \mathbf{F} = 0 - 0 + 0 = 0$$

Thus, the curl of the vector field $\mathbf{F}$ is 0.

You are making right steps, you are now going to look at Multiple integrals.

4 Multiple Integrals

Definition: Integrals involving more than one variable.

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/SKD4A6mi9l0>



Introduction to double and triple integrals

5 Double Integrals

Video Visit the URL below to view a video:

<https://www.youtube.com/embed/R-SS8pRrvDo>



Evaluating double integrals

They are used to compute areas and volumes over a 2D region R :

$$\iint_R f(x, y) dA$$

Example 8. • *Function:* $f(x, y) = 1$

• *Region:* Unit square $[0, 1] \times [0, 1]$

• *Integrat:*

$$\iint_{[0,1] \times [0,1]} 1 dx dy = \int_0^1 \int_0^1 1 dx dy = 1$$

Example 9. Compute the double integral of $f(x, y) = x + y$ over the rectangle $0 \leq x \leq 2$ and $0 \leq y \leq 3$.

Solution:

$$\begin{aligned} \int_0^2 \int_0^3 (x + y) dy dx &= \int_0^2 \left(x \cdot y + \frac{y^2}{2} \Big|_0^3 \right) dx \\ &= \int_0^2 \left(3x + \frac{9}{2} \right) dx \\ &= \left[\frac{3x^2}{2} + \frac{9x}{2} \Big|_0^2 \right] \\ &= \frac{6 + 9}{2} = 15 \end{aligned}$$

6 Triple Integrals

Video Visit the URL below to view a video:

https://www.youtube.com/embed/ffS_C4IvCR8



Evaluating triple integrals

They are used to compute volumes and cumulative effects over a 3D region V :

$$\iiint_V f(x, y, z) dV$$

Example 10. • *Function:* $f(x, y, z) = z$

• *Region:* Unit cube $[0, 1] \times [0, 1] \times [0, 1]$

- **Integral:**

$$\begin{aligned} & \iiint_{[0,1] \times [0,1] \times [0,1]} z dx dy dz \\ &= \int_0^1 \int_0^1 \int_0^1 z dx dy dz \\ &= \int_0^1 \int_0^1 \left[\frac{z^2}{2} \right]_0^1 dy dx \\ &= \int_0^1 \int_0^1 \frac{1}{2} dy dx = \frac{1}{2} \end{aligned}$$

7 Applications in AI Problems

In machine learning, the gradient vector indicates the direction of the steepest ascent of a function (e.g., the loss function). To optimize the function (usually minimizing it), algorithms like gradient descent move in the direction opposite to the gradient. This helps in finding the minimum of the function, which corresponds to the optimal parameters for the model. In training a neural network, the gradient descent uses partial derivatives to update the weights of the neural network. By calculating the gradient of the loss function with respect to each weight (using partial derivatives), the algorithm determines the direction and magnitude to adjust the weights to minimize the loss function.

8 Gradient Descent for a Loss Function

Example 11. *In machine learning, gradient descent is used to minimize the loss function of a model. Consider a simple loss function given by:*

$$L(w) = w^2 - 4w + 4$$

where w is a parameter of the model. Find the gradient of this loss function and determine the value of w that minimizes the loss.

Solution

- *Compute the Gradient:* The loss function $L(w)$ is a scalar function of w . To find the gradient, we compute the derivative of $L(w)$ with respect to w :

$$\nabla L(w) = \frac{dL}{dw}$$

Calculate:

$$\frac{dL}{dw} = \frac{d}{dw} (w^2 - 4w + 4) = 2w - 4$$

Thus:

$$\nabla L(w) = 2w - 4$$

- *Find the Minimum:* To minimize the loss function, we set the gradient to zero:

$$2w - 4 = 0$$

Solve for w :

$$2w = 4 \implies w = 2$$

Therefore, the value of w that minimizes the loss function is $w = 2$.

- *Verify the Minimum:*

To confirm that $w = 2$ is a minimum, we check the second derivative:

$$\frac{d^2L}{dw^2} = \frac{d}{dw} (2w - 4) = 2$$

Since $\frac{d^2L}{dw^2} > 0$, the function is concave up at $w = 2$, confirming a local minimum.

Answer: The parameter w that minimizes the loss function $L(w)$ is $w = 2$.

9 Gradient of a Cost Function in Neural Networks

Example 12. Consider a simple neural network with a cost function given by:

$$C(w) = \frac{1}{2}(y - Xw)^\top (y - Xw)$$

where $\mathbf{X}$ is an $m \times n$ matrix of input features, w is an $n \times 1$ vector of weights, and y is an $m \times 1$ vector of target values. Find the gradient of the cost function with respect to the weight vector w .

Solution:

1. Define the Cost Function:

The cost function can be expressed as:

$$C(w) = \frac{1}{2}(y - Xw)^\top (y - Xw)$$

2. Compute the Gradient: To find the gradient of $C(w)$ with respect to w , we first need to expand the cost function:

$$C(\mathbf{w}) = \frac{1}{2} [\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - (\mathbf{X}\mathbf{w})^\top \mathbf{y} + (\mathbf{X}\mathbf{w})^\top (\mathbf{X}\mathbf{w})]$$

3. Simplify the terms:

$$C(w) = \frac{1}{2} [\mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{w} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}]$$

4. Compute the gradient with respect to w :

$$\nabla_{\mathbf{w}} C(\mathbf{w}) = -\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w})$$